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Ascent trajectory design of small-lift launch vehicle using hierarchical optimization

Yaru Zheng [a], Xiaoyu Fu [b], Ming Xu [b,*], Qinglong Li [b], Mingpei Lin [b]

[a] The 9th Designing, CASIC, Wuhan 430040, China
[b] School of Astronautics, Beihang University, Beijing 100191, China

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ABSTRACT

An integrated algorithm to design the optimal ascent trajectory of a target Sun-synchronous orbit (SSO) for small-lift launch vehicles is studied in this paper. A general problem description is provided, including the specific pitch law designed for an ascent trajectory to SSO and the definitions of design variables and equality/inequality constraints. Then an integrated optimization scheme is introduced to solve this problem. In order to improve its efficiency, a preliminary determination method is to analytically generate two sensitive design variables related with the coast arc. With the help of the two design variables, an initial combination of design variables can be obtained by solving separately the optimization problems of two powered phases. Based on the predetermination method, an efficient integrated algorithm using hierarchical optimization is proposed. A simulation based on a type of small-lift Electron Rocket and Rutherford engine is performed using this hierarchical optimization algorithm and the feasibility and efficiency of this algorithm is evaluated. These systematic investigations on all the possible injected orbits give a full detail of launching capability for Electron Rocket.

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1. Introduction

Safe, dependable, and cost-effective access to orbits near the Earth is necessary for all spacefaring nations' endeavors. The small-lift launch vehicle, which is capable of lifting up to 2000 kg of payload into the low Earth orbit (LEO) [1], has been performing an important role in the safe and reliable launch since the middle of last century. Great advances that have been made in propulsion technologies improve cost-effectiveness and flexibility in space tasks, such as the launch of microsatellites [2]. The small-lift launch vehicles will attract more attention in the future space tasks used in long-time transportation. Electron Rocket is one of the small-lift launch vehicle, which has attracted lots of attention in aerospace society and has been applied into practice such as the Electron launched by Rocket Lab.

The preliminary design of ascent trajectories for a given launch vehicle can demonstrate the feasibility of a mission with this vehicle. Practical results, including analytic solutions, empirical relationships and feedback control, were obtained in the simplified models of launch vehicle [3,4]. With the improvements of computer technology, numerical method and analytic/numerical methods were widely employed in trajectory design problems. The corresponding algorithms can be categorized as indirect algorithms, Liang et al. gives a sliding mode control on the VTVL launch vehicle, Cheng et al. has tested the impact-time-control, Xu et al. offers a multi-constrained method for solid propellant launch vehicle [5][6][7]. Coverstone-Carrol et al. have proposed a direct algorithm using different inclusion concepts, while Cheng et al. study efficient ascent trajectory optimization using convex models based on the Newton-Kantorovich/pseudospectral approach [8][9].

Researches on trajectory optimization of multistage launch vehicles also rely on different numerical methods. Beltracchi [10] proposed a decomposition algorithm to solve a maximum payload problem with upper-stage sizing and a fixed parking orbit perigee altitude. By solving booster and upper-stage problems separately, the algorithm can be applied to acquire an approximation of the all-up performance of a launch vehicle. Wang [11] et al. gave a robust optimization using polynomial chaos and convex optimization. Gath and Calise [12] developed and improved a hybrid analytical/numerical algorithm for optimization of launch vehicle trajectories. Their improvement also made the algorithm applicable to real-time guidance of advanced launch vehicle configuration. Ross et al. [13] proposed a fast and accurate approach for computing optimal trajectories for multi-stage launch vehicles, the computational technique of which was mainly based on the pseudospectral knotting method. Dukeman and Hill [14] developed a sim-

* Corresponding author.
E-mail address: xuming@buaa.edu.cn (M. Xu).

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plified ascent trajectory optimization algorithm, in which the design space was reduced to two free parameters, an initial pitch rate and the launch azimuth. Jeshvaghani et al. [15] applied Chebyshev wavelets to determine an optimal trajectory of a multiple stage satellite launch vehicle. The previously cited works [5,7,12] utilize indirect algorithms and resulting drawbacks are obvious, including an excessive dependence on the starting guess, considerable effort for deriving the analytic conditions for optimality. Especially to circumvent the excessive reliance on the initial guess, the homotopic method was proposed [12,16] and global optimization methods, such as genetic algorithm (GA) [17] and particle swarm optimization (PSO) [18,19], was introduced in the optimal trajectory design of multistage launch vehicles. However, in the former occasions where homotropy is used, further complexity from this method is induced and the efficiency of this method is conditional. In the later occasions where the global optimization method is applied to obviate the selection of initial value, considerable computation is inevitable and the slow rate of convergence and even nonconvergence is frequently expected, especially in the optimization of trajectory including coast arcs, such as the ascent trajectory for Sun-synchronous orbit (SSO).

The work in this paper is concerned with an efficient and reliable algorithm incorporating both existing empirical relationships and global optimization method to fulfill the ascent trajectory design for an electronic rocket which is a small-lift launch vehicle with given characteristics. The approach described in this work is applied to a two-stage rocket, Electron, of which the three-dimensional ascent trajectory for SSO includes a coast arc between the first-time second engine cut-off (1st SECO) and the second-time second engine cut-off (2nd SECO). Thus, an integrated algorithm using hierarchical optimization is proposed. One of the contributions in this paper is to firstly design the ascent trajectory systemically for the new type of Electron rockets. These systematic investigations on all the possible injected orbits give a full detail of launching capability for Electron Rocket. Furthermore, another contribution is that the initial mass of payload and left mass are derived analytically based on the idea approximating burnout velocity of rocket within a short period. The third contribution of this paper is to build an integrated two-level optimization algorithm for nine design variables of Electron Rocket. Compared with the traditional optimization, the computation time has been reduced which offers a feasible method to practice. Ballistic results from this hierarchical optimization algorithm are proved to be reasonable.

2. Problem definition of optimal ascent with coast arc for electronic rockets

This research addresses the design of optimal ascent trajectory of a small-lift launch vehicle for a target SSO. As a typical kind of ascent trajectory with a coast arc, the ascent trajectory for an SSO includes a long coast arc, which guarantees the injection altitude of a payload. In comparison with medium-lift and heavy-lift launch vehicles, a small-lift launch vehicle is not equipped with boosters and typically designed to be two-stage, which makes its modeling relatively simple. In addition, the diameter variation along the axial direction of the launch vehicle is relatively small, which allows a relatively convenient determination of the lift and drag coefficients.

2.1. Equation of motion in the launch position reference frame

Dynamics of the center of mass (COM) is established to calculate the trajectory of a small-lift launch vehicle flying over the spherical, rotating Earth. The motion of the launch vehicle's COM is described in a rotational reference frame that is rigidly attached to the Earth at launch position. The equation of motion in the launch position reference frame is given in a vector form as follows:

$$m\boldsymbol{a}_r = \boldsymbol{P} + \boldsymbol{A} + m\boldsymbol{g} - m\boldsymbol{a}_e - m\boldsymbol{a}_c \tag{1}$$

where m denotes the mass of the launch vehicle, $\boldsymbol{a}_\mathrm{r}$ is the acceleration of the COM, $\boldsymbol{P}$ is the thrust of rocket engines, $\boldsymbol{A}$ is the aerodynamic force exerted on the vehicle, $\boldsymbol{g}$ is gravitational acceleration, $\boldsymbol{a}_\mathrm{e}$ is the centrifugal acceleration of the COM, $\boldsymbol{a}_\mathrm{c}$ is the Coriolis acceleration related to the motion of the COM. For the convenience of expression, a sign of (t) is omitted in the denotations of variables in Eq. (1). However, the time-variant properties of variables are still taken into account to make the modeling in Eq. (1) more reasonable. The thrust-atmospheric effect is considered in the thrust $\boldsymbol{P}$. The aerodynamic force $\boldsymbol{A}$ varies due to the change of the atmospheric density and vehicle's velocity. The J_2 perturbation effect is incorporated in the linear gravity approximation of gravitational acceleration $\boldsymbol{g}$. In addition, variables mentioned above are not defined directly in launch position coordinate system. The thrust $\boldsymbol{P} = [P_x\ P_y\ P_z]^\mathrm{T}$ is defined in the vehicle-body reference frame. The aerodynamic force $\boldsymbol{A} = [A_x\ A_y\ A_z]^\mathrm{T}$ is defined in the airflow reference frame. Thus, Eq. (1) can be expanded as follows:

$$\begin{bmatrix} \dfrac{dv_x}{dt} \\ \dfrac{dv_y}{dt} \\ \dfrac{dv_z}{dt} \end{bmatrix} = \frac{1}{m}\boldsymbol{L}_{lb}\left(\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} + \boldsymbol{L}_{ba}\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}\right) + \{\boldsymbol{g}\}_l - 2\{\boldsymbol{\omega}_e\}_l \times \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} - \{\boldsymbol{\omega}_e\}_l \times \{\boldsymbol{\omega}_e\}_l \times \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{2}$$

where $[x\ y\ z]^\mathrm{T}$ and $[v_x\ v_y\ v_z]^\mathrm{T}$ denotes the position vector and velocity vector of the launch vehicle respectively. $\boldsymbol{L}_{lb}$ is the transformation matrix from the vehicle-body reference frame to the launch position reference frame, $\boldsymbol{L}_{ba}$ is the transformation matrix from the airflow reference frame to the vehicle reference frame. $\{\boldsymbol{g}\}_l$ and $\{\boldsymbol{\omega}_e\}_l$ respectively denote the projections of gravitational acceleration and the Earth's rotational angular velocity in the launch position reference frame. The definition of the different reference frames mentioned above and their related transformation matrices can be found in Ref. [20].

2.2. Pitch law of the flight ascending to SSO

The pitch law during a specific flight has a direct influence on the carrying capacity of a launch vehicle with given characteristics. The design of pitch law during the flight along an ascent trajectory is closely related to the launch sequence of a specific task, especially in the consideration of the existence of coast arcs. During powered phases, a multi-segment linear pitch law can be designed, with the pitch rates vary within the limitation of machine in the control system. However, at the beginning of the launch, two successive processes are relatively fixed, which include vertical ascending section and pitch-over section. A launch vehicle is restricted to fly vertically in the former section and the determination of the pitch law in the latter section relies on the empirical formula. During a Hohmann transfer outside the atmosphere, active control along the corresponding coast arc is not necessary. A pitch law directly determined by gravity can meet the requirement upon the flight along the coast arc. Based on the discussions above, in the consideration of a small-lift launch vehicle, the pitch law of

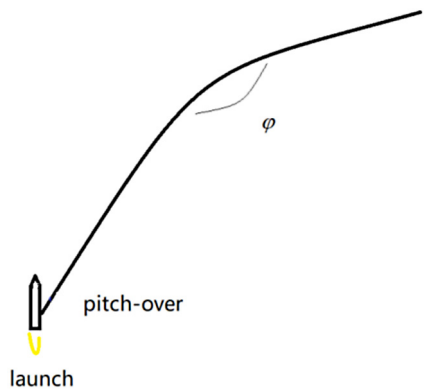

Fig. 1. Pitch-over maneuver of the ascent trajectory.

the flight ascending to a SSO is separated into five successive sections, which is expressed as follows:

$$\varphi(t) = \begin{cases} \frac{\pi}{2} \\ 0 \le t < t_{PO-start} \\ \theta(t) + \alpha(t) + \omega_z t \\ t_{PO-start} \le t < t_{PO-end} \\ \varphi(t_i) - \dot{\varphi}_i (t - t_i) \\ t_{PO-end} \le t_i \le t < t_{SECO-1st},\ i = 1, 2, ..., n \\ \varphi(t_{SECO-1st}) - \int_{t_{SECO-1st}}^{t} \dot{\varphi}_{gravity}(t)\, dt \\ t_{SECO-1st} \le t < t_{reboot} \\ \varphi(t_{reboot}) - \dot{\varphi}_{n+1}(t - t_{reboot}) \\ t_{reboot} \le t \le t_{SECO-2nd} \end{cases} \tag{3}$$

where as φ is called pitch-over angle. After the first ignition, the rocket need change its orientation lined up with the launch azimuth and a pitch-over maneuver is done to achieve a downrange velocity vector to achieve an orbital velocity or down range displacement, where the angle changing in the flight path is called pitch-over angle shown in Fig. 1.

In the first section, the vertical ascending section, the start time for pitch-over $t_{PO\text{-}start}$ directly depends on the initial thrust-to-weight ratio N_0 of a launch vehicle and can be approximated by the empirical relationship [21] as $t_{PO-start} = \sqrt{\frac{40}{N_0 - 1}}$. In the section of pitch-over ($t_{PO\text{-}start} \sim t_{PO\text{-}end}$), the Earth's rotational angular velocity $\boldsymbol{\omega}_z t$ is taken into account apart from the flight path angle $\theta(t)$ and the angle of attack $\alpha(t)$. Based on existing empirical relationship [21], the attack angle of a small-lift launch vehicle varies exponentially and can be expressed as follows:

$$\alpha(t) = -4\alpha_{\max}\left[1 - e^{\kappa(t_{PO-start} - t)}\right] \tag{4}$$

where κ denotes the coefficient of pitch-over, α_{max} denotes the maximum of attack angle. The two parameters are not fixed and remain to be optimized. The end time of this section $t_{PO\text{-}end}$ is selected to be at the time of main engine cut-off (MECO). After the flight of the first two sections, the upper stage of a launch vehicle is supposed to achieve a sufficiently large velocity and reach an altitude over 100 km.

In the third section ($t_{PO\text{-}end} \sim t_{SECO\text{-}1st}$) where the rocket engine of upper stage almost finishes working, the corresponding pitch law is divided into n successive linear segments. The derivative of the pitch angle in the ith segment is denoted as $\dot{\varphi}_i$ $(i = 1, 2, \cdots, n)$, which remains to be optimized. Considering a launch vehicle with fixed total mass of fuel and a rocket engine of constant mass flow rate, the time of second engine first cut-off $t_{SECO\text{-}1st}$ is indirectly determined by the fuel left for reboot. In the following section ($t_{SECO\text{-}1st} \sim t_{rebost}$) which includes a coast arc, the derivative of the pitch angle is directly determined by gravity and thus denoted as $\dot{\varphi}_{gravity}$. For no active control, the pitch angle in this section can be calculated directly from the integration results of an ascent trajectory. In the last section ($t_{rebost} \sim t_{SECO\text{-}2nd}$), active control is retaken and the derivative of the pitch angle in this section is denoted as $\dot{\varphi}_{n+1}$.

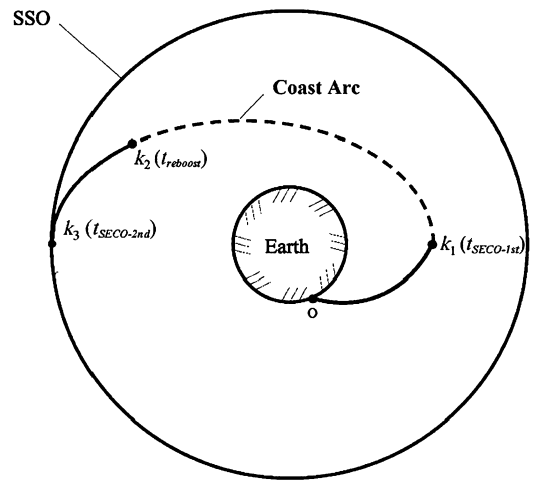

Fig. 2. An ascent trajectory to SSO which includes a coast arc.

2.3. Problem description of the optimal ascent trajectory

An ascent trajectory including a coast arc can efficiently increase the injection altitude, in comparison with a trajectory containing only thrust phases. Thus, the following optimal ascent trajectory is composed of three successive parts, which include a coast arc between the second engine first cut-off and reboot. A schematic representation of this trajectory is shown in Fig. 1. Although it has been projected in a two-dimensional plane, the actual nominal trajectory is designed using the three-dimensional model discussed in Section 2.1. The first thrust phase of this trajectory is from the launch position on the Earth to k_1, the injection position of a predesigned transfer orbit. The second thrust phase begins at position k_2, which is near the injection position k_3 of a target SSO. As the altitude of the apogee of the transfer orbit is designed to be consistent with the altitude of the target SSO, the second thrust phase mostly renders sufficient velocity increment to a payload. A detailed expression of the design problem about the ascent trajectory shown in Fig. 2 follows a general standard form of a nonlinear optimization problem, which is habitually described as

$$\begin{cases} \min\limits_{\boldsymbol{x}} f(\boldsymbol{x}) = \min\limits_{\boldsymbol{x}} [M(\boldsymbol{x})] \\ \quad s.t.\ h_j(\boldsymbol{x}) = 0, \\ \quad\quad g_i(\boldsymbol{x}) \le 0, \\ \boldsymbol{x} = [x_1, x_2, \cdots x_n]^T \end{cases} \tag{5}$$

where $\boldsymbol{x} = [x_1, x_2, \ldots, x_n]^{\mathrm{T}}$ denotes the bounded design variables; $\boldsymbol{f}(x)$ represents the objective function; g and h represent inequality constraints and equality constraints respectively.

In this case, the pitch law in the third section ($t_{PO\text{-}end} \sim t_{SECO\text{-}1st}$) is divided into two successive linear segments. Although more segments can improve the continuity between derivatives of pitch angle, two segments are qualified to satisfy the design in this section [22]. Thus, nine design variables are selected for this optimization problem, including 1) x_1: launch azimuth, 2) x_2: maximum of attack angle α_{max}, 3) x_3: coefficient of pitch-over κ, 4) x_4: derivative of the pitch angle in the first segment $\dot{\varphi}_1$, 5) x_5: derivative of the pitch angle in the second segment $\dot{\varphi}_2$, 6) x_6: derivative

of the pitch angle in the last section $\dot{\varphi}_3$, 7) x_7: the time of reboot of second-stage engine, 8) x_8: the mass of propellant left for reboot m_{left}; 9) x_9: the mass of payload $m_{payload}$. The objective function $f(x)$ is the maximum mass of the payload injected into a target SSO. During the flight inside the atmosphere, the launch vehicle is constrained by 1) maximum dynamic pressure encountered, 2) maximum axial overload encountered, and 3) maximum attack angle. The above three factors are transformed into the inequality constraints g in this problem. The equality constraints h in this problem come from the target injection positions at k_1 and k_3. There are eight equality constraints in this problem, including orbital altitude, orbital inclination, orbital eccentricity, and flight path angle at both target injection positions.

3. Hierarchical optimization algorithm for the design of ascent trajectory

3.1. Integrated optimization scheme based on global optimization method

As demonstrated in Section 2.3, the design problem of optimal ascent trajectory to SSO is composed of nine design variables, three inequality constraints and eight equality constraints. Considering the complexity of this problem, decomposition approach has been proposed to search a near-optimal solution for a multistage launch vehicle [10,22,23]. However, a more direct approach to this problem is to design an algorithm based on global optimization method. As multimodality has been analytically or numerically proved to exist in many practical optimization problems, local optimization methods are commonly ineffective to find a global optimum in these cases. Instead, global reliability and local refinement are considered simultaneously in a global optimization method [24], which can guarantee the acquirement of a global optimum theoretically. Different global optimization methods, including Genetic Algorithm (GA), Participle Swarm Optimization (PSO), and Differential Evolution (DE) method, have been developed and widely adopted in different problems [25–27]. A comparative analysis about the performance of global optimization of launch vehicle further demonstrates the feasibility of the developed global optimization architecture and algorithms [28]. In consideration of the structure of general global optimization method and the definition of this ascent trajectory design problem, an integrated optimization scheme is designed and illustrated in the flow chart of Fig. 3.

As shown in Fig. 3, the integrated optimization scheme of ascent trajectory design is straightforward and can be realized by different global optimization methods. The generation of initial set of design variables can rely on a blind research method or a predetermined range based on practice constraints. The techniques to screen design variables are different among different methods, such as the crossover, mutation, and the corresponding fitness function used in the Genetic Algorithm. Thus, a wide-ranging search capacity with adjustable precision is possessed by this integrated scheme and a global optimum can be obtained theoretically.

However, the simulation result of this integrated optimization algorithm using a blind research method at the beginning is far from satisfaction. Taking the GA as an example, a larger population size means a larger possibility of convergence after the mutation and selection of limited generations. Meanwhile, a larger population size is also tantamount to a heavier computation burden, which should be circumvented in the phase of designing a nominal trajectory. In addition, as demonstrated by Lu et al. [5], the presence of coast arcs renders this optimal control problem more sensitive and brings more trouble to achieve convergence of the related algorithm. In this case, the last three design variables, including the time of reboot of second-stage engine, the mass of propellant left for reboot m_{left}, and the mass of payload $m_{payload}$,

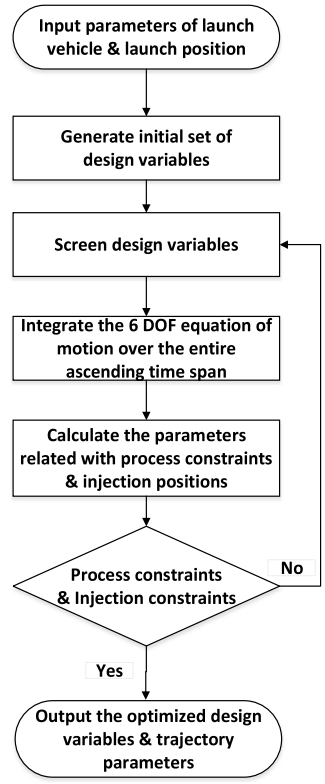

Fig. 3. Integrated optimization scheme based on global optimization method.

are much more sensitive in comparison with the rest six design variables. Instead of a random evaluation, the preliminary determination of these three design variables which are closely related to the coast arc should be further researched.

3.2. Integrated algorithm using hierarchical optimization

Based on the analytic approach to preliminarily determine the two design variables m_{left} and $m_{payload}$, an algorithm using hierarchical optimization is proposed to improve the efficiency and accuracy of integrated optimization scheme discussed in Section 3.1. Instead of using initial design variables generated randomly from a search space, a logical selection of initial design variables is performed in the lower level of this hierarchical optimization algorithm. The initial guess of m_{left} and $m_{payload}$ can not only be used in the initial combination of nine design variables, but also used to pre-determine the rest seven design variables. In consideration of the two powered phases, the rest initial design variables can be generated by separately solving two optimization problems related with the two powered phases. The methods to solve these two separated optimization problems have been widely studied and proved to be effective [19]. For the first problem corresponding to the first powered phase, initial values of five design variables can be determined with the value of m_{left} and $m_{payload}$, include 1) launch azimuth, 2) maximum of attack angle $\alpha_{\max}$, 3) coefficient of pitch-over κ, 4) derivative of the pitch angle in the first segment $\dot{\varphi}_1$, and 5) derivative of the pitch angle in the second segment $\dot{\varphi}_2$. Derivative of the pitch angle in the last section $\dot{\varphi}_3$ can be determined from the problem related with the second powered phase. This algorithm using hierarchical optimization is illustrated in Fig. 4.

To avoid the optimizing iterations diverged by more sensitive variables, the analytical solutions to the mass of propellant left for reboot m_{left} and the mass of payload $m_{payload}$ derived in this paper, are equipped as the inner level of integrated optimization to

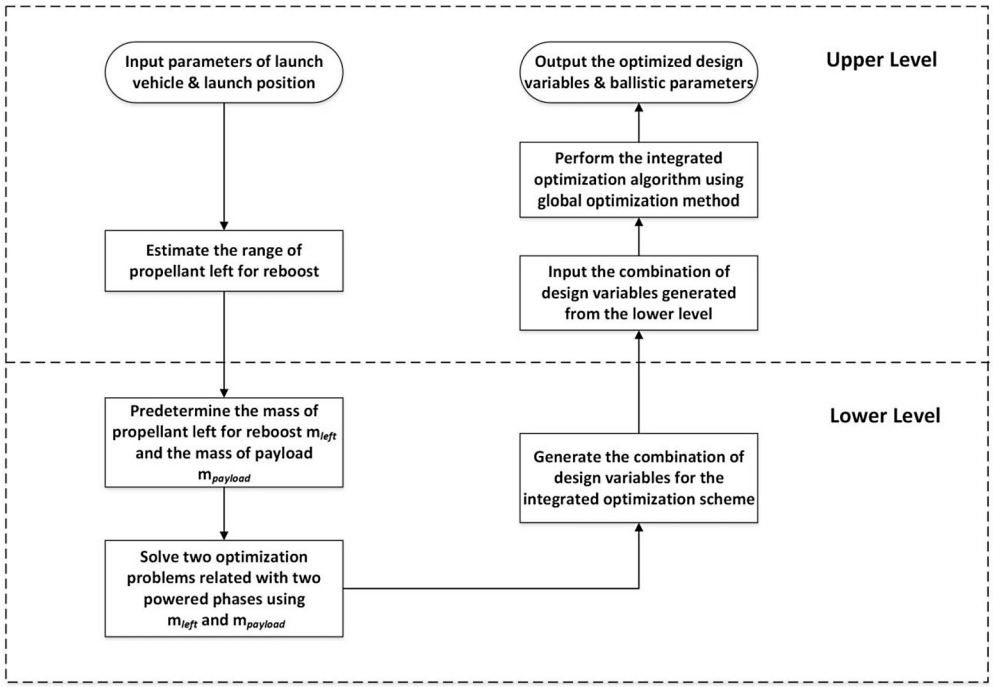

Fig. 4. Integrated algorithm using hierarchical optimization.

provide initial values for the further optimization by the following outer level. Thus, the rest seven design variables generated from the predetermined m_{left} and $m_{payload}$ also approach their final optimums. The combination of design variables above can guarantee the convergence and efficiency of this integrated hierarchical optimization algorithm. In addition, to avoid the iterations slowed by the other seven variables without any analytical gradient information, random methods are preferred to be the outer level of integrated algorithm, such as Genetic Algorithm (GA), Participle Swarm Optimization (PSO), Differential Evolution method (DE). Therefore, the method used in this paper can be a potential optimizer embedded in the outer level to widely suitable for more optimization scenarios. Without loss of generality, we embed a common GA into the outer level to product optimal trajectories for specified scenarios in Section 4.

3.3. Analytic initial guess of design variables m_{left} and $m_{payload}$

Based on the analysis in Section 3.1, the relationship between the mass of propellant left for reboot m_{left} and the mass of payload $m_{payload}$ is studied analytically in this section. The result from this relationship will be used to perform an initial guess of design variables. As shown in Fig. 1, an entire ascent trajectory to an SSO includes two powered phases: trajectory from the launch position O to the injection position k_1 of a predesigned transfer orbit, and trajectory from k_2, the position of reboot to the injection position k_3 of a target SSO. The flight in former phase is similar with a flight whose target orbit is a low-Earth orbit (LEO). The difference lies in a fact that the nominal payload at position k_1 should be taken into account both the actual payload (e.g. satellite) and the propellant left for reboot. As the carrying capacity of a launch vehicle to a target LEO is relatively fixed, a negative correlation is able to be anticipated between m_{left} and $m_{payload}$ in the flight from position O to position k_1, as the mass of propellant left for reboot is taking up the portion of the mass. On the contrary, in the flight from position k_2 to position k_3, a positive correlation can be anticipated between m_{left} and $m_{payload}$, as a larger velocity increment can be provided a larger amount of fuel left for reboot. Thus, an intersection is supposed to be obtained from the relations of m_{left} and $m_{payload}$ in the two powered phases. The value at the intersection can be leveraged to perform an initial guess of m_{left} and $m_{payload}$. Based on the qualitative analysis above, a detailed analytic approach is presented.

This analytic approach is inspired and derived from the work of White [4], where an estimation method is provided and demonstrated to approximate the burnout velocity of a launch vehicle within a short period of time. Basically following his framework [4], adjustments have been made to the analytic approach for the case of an ascent trajectory to a SSO. Theoretically, for a small-lift launch vehicle, the injection velocity v_{inj} of its payload should be no larger than the vehicle's final burnout velocity $v_{BO.final}$, which can be expressed as

$$v_{BO.final} = \sum_{i=1}^{N} \left(v_I^{(i)} - \Delta v_L^{(i)} \right) \tag{6}$$

where N denotes the number of stages of a launch vehicle, $v_I^{(i)}$ denotes the ideal velocity of the ith stage of a launch vehicle, $\Delta v_L^{(i)}$ denotes the velocity loss of the ith stage. The ideal velocity of the ith stage is determined by the following Tsiolkovsky equation:

$$v_I^{(i)} = g_0 Isp_{vac}^{(i)} \ln \frac{W_O^{(i)}}{W_{BO}^{(i)}} \quad i = 1, 2, 3, \ldots \tag{7}$$

where g_0 is the standard gravity $= 9.80665$ m/s^2, $Isp_{vac}^{(i)}$ is the vacuum specific impulse of the rocket engine in the ith stage. $W_O^{(i)}$ and $W_{BO}^{(i)}$ denote respectively the initial weight and weight at burnout of the ith stage. The expression of the velocity loss $\Delta v_L^{(i)}$ is provided in the form of a piecewise function:

$$\Delta v_L^{(i)} = \begin{cases} \Delta v_{L_g}^{(i)} + \Delta v_{L_D} + \Delta v_{L_T} + \Delta v_{L_r} & \text{if } i = 1 \\ K_{multi} \Delta v_{L_g}^{(i)} & \text{if } i = 2, 3, \ldots \end{cases} \tag{8}$$

As shown in Eq. (8), the approximation of the velocity loss $\Delta v_L^{(i)}$ is divided into two cases considering different atmospheric effects during the flight of first stage and upper stage. During the flight of the first stage, the velocity loss $\Delta v_L^{(1)}$ is mainly composed

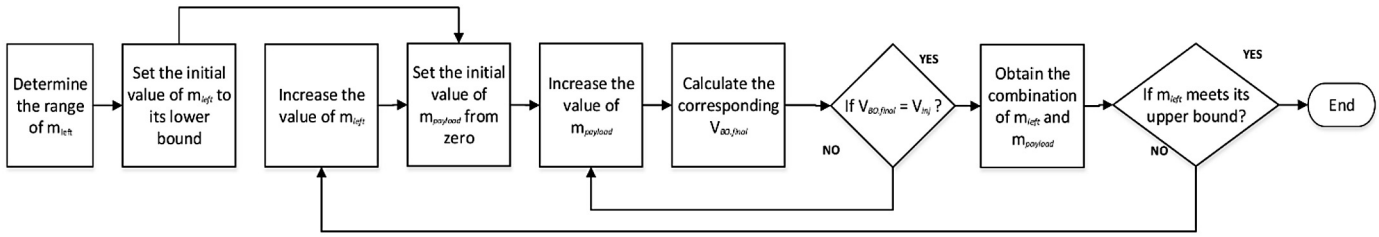

Fig. 5. Numerical search method to solve the relationship between m_{left} and $m_{payload}$ in the first powered phase.

of four terms: gravitational loss $\Delta v_{L_g}^{(1)}$, drag loss Δv_{L_D}, thrust-atmospheric loss Δv_{L_T}, and velocity loss arising from the Earth's rotation Δv_{L_r}. The gravitational loss of the first stage $\Delta v_{L_g}^{(1)}$ is expressed in the following equation,

$$\Delta v_{L_g}^{(1)} = (g_0 t_{B_{stage1}} - K_{gg})[1 - K_g(1 - \frac{W_{BO}^{(1)}}{W_O^{(1)}})\left(\frac{\beta_{BO_{stage1}}}{90}\right)^2] \quad (9)$$

where $t_{B_{stage1}}$ is the first-stage burning time, $\beta_{BO_{stage1}}$ is the first-stage relative flight path angle at burnout with respect to local vertical (unit: deg), r is the first-stage mass ratio which satisfies $r = W_O^{(1)}/W_{BO}^{(1)}$. K_{gg} and K_g are two empirical constants which are used to correct the gravitational loss. K_{gg} is used to correct the overestimated loss induced by the term $g_0 t_{B_{stage1}}$ and K_g is a function of initial thrust-to-weight ratio $T/W_O^{(1)}$, first-stage vacuum specific impulse $Isp_{vac}^{(1)}$, and first-stage mass ratio r. The drag loss Δv_{L_D} of the first stage is determined by the following empirical equation,

$$\Delta v_{L_D} = K_D \frac{C_{D_{PD}} A}{W_O^{(1)}} \quad (10)$$

where A is the reference area of a launch vehicle, K_D and $C_{D_{PD}}$ are two empirical constants. K_D denotes the drag loss coefficient which is a function of initial thrust-to-weight ratio $T/W_O^{(1)}$, first-stage vacuum specific impulse $Isp_{vac}^{(1)}$, and flight path angle of the first stage at burnout $\beta_{BO_{stage1}}$. $C_{D_{PD}}$ is the drag coefficient at maximum drag force during the trajectory. The thrust-atmospheric loss Δv_{L_T} is caused by the reduction of vacuum thrust level by atmospheric pressure, especially during the first-stage flight through the atmosphere. For the conical and DeLaval type nozzles, the corresponding thrust-atmospheric loss can be approximated by the following equation:

$$\Delta v_{L_T} = K_a \quad (11)$$

where K_a is an empirical constant and K_a is a function of initial thrust-to-weight ratio $T/W_O^{(1)}$. The empirical constants mentioned above can be determined by referring to empirical relationships shown in the figures of Ref. [4]. The velocity loss Δv_{L_r}, which arises from the Earth's rotation is common in the tasks related with SSO, as a westward launch inevitably decreases the final burnout velocity. Δv_{L_r} can be expressed as

$$\Delta v_{L_r} = v_{BO_{stage1}}^* - \sqrt{{v_{BO_{stage1}}^*}^2 + {v_{rot}}^2 - 2 v_{BO_{stage1}}^* v_{rot} \sin\beta_{BO_{stage1}}} \quad (12)$$

where $v_{BO_{stage1}}^*$, which satisfies $v_{BO_{stage1}}^* = v_I^{(1)} - \left(\Delta v_{L_g}^{(1)} + \Delta v_{L_D} + \Delta v_{L_T}\right)$, is the first-stage burnout velocity in a non-inertial coordinate system. v_{rot} is the rotational velocity of the launch site.

In comparison with the complex expression of the first-stage velocity loss, the expression of upper-stage velocity loss is relatively simpler. The upper-stage velocity loss is large due to gravity and angle of attack. Within the two factors, the gravity loss is the more significant than the latter. Thus, the upper-stage loss can be approximated as

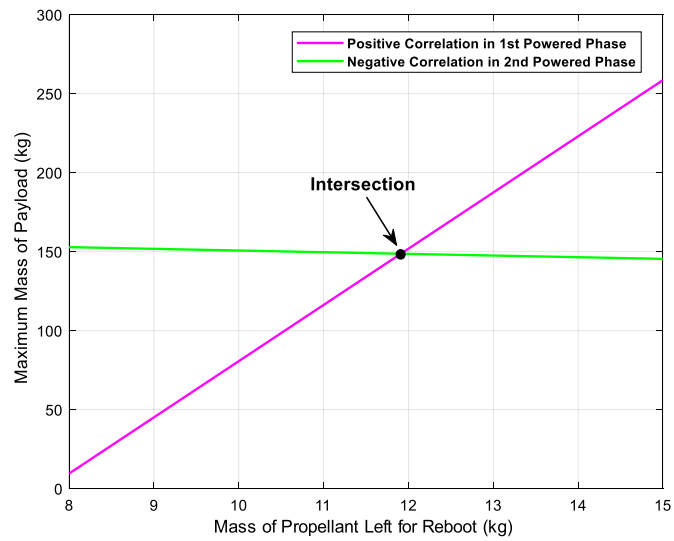

Fig. 6. Correlations between m_{left} and $m_{payload}$ in two powered phases. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

$$\Delta v_L^{(i)} = K_{multi} \Delta v_{L_g}^{(i)} = K_{multi} g_0 t_B^{(i)} \cos\overline{\beta}^{(i)} \quad i = 2, 3, \ldots \quad (13)$$

where K_{multi} is a correction coefficient in consideration of the multiple factors apart from the gravity, $t_B^{(i)}$ is the burning time of the ith stage, $\overline{\beta}^{(i)}$ is the average flight path angle of the ith stage. An empirical curve of $\cos\overline{\beta}^{(i)}$ is shown in Ref. [4]. As the determination of ideal velocity, velocity loss, and related variables are directly or indirectly related with the initial weight and burnout weight, a numerical search method can be designed to figure out the relationship between m_{left} and $m_{payload}$ in the flight from position o to position k_1. Based on the analytic equations from Eq. (6) to Eq. (13), this numerical search method is illustrated in the flow chart shown in Fig. 5.

In comparison with the indirect determination method of the relationship between m_{left} and $m_{payload}$ in the first powered phase, the relationship of the two design variables from position k_2 to position k_3 can be solved directly from the Tsiolkovsky equation by an approximation. As the position of k_2 is close to the apogee of the transfer orbit, the velocity increment after reboot Δv_{reboot} can be approximated by $\Delta v_{reboot} = v_{inj} - v_{trans.apo}$, where $v_{trans.apo}$ denotes the velocity at apogee of transfer orbit. Thus, the relationship between m_{left} and $m_{payload}$ in the second powered phase can be expressed in the following linear form as

$$m_{payload} = -m_{dry}^{(2)} + \frac{1}{r-1} m_{left} \quad (14)$$

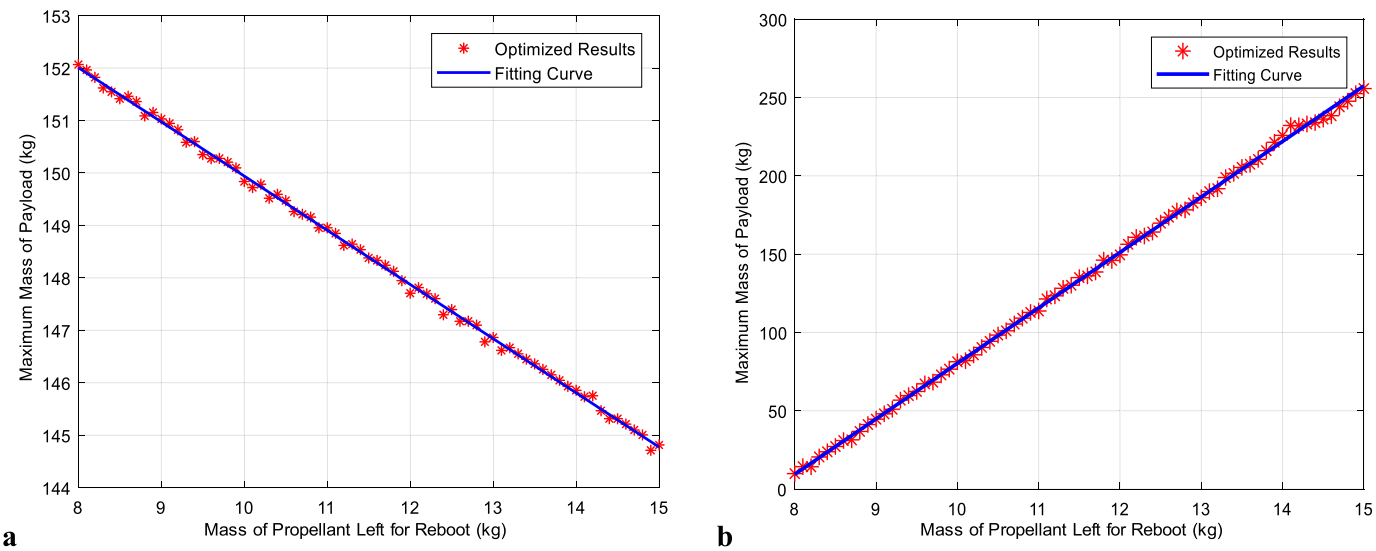

Fig. 7. Optimized results of two optimization problems and corresponding fitting curves: **a** optimized results and fitting curve in the first power phase, **b** optimized results and fitting curve in the second power phase.

where $m_{dry}^{(2)}$ denotes the structural mass of the second stage, and r is expressed as

$$r = \exp\left(\frac{\Delta v_{reboot}}{g_0 Isp_{vac}^{(2)}}\right). \tag{15}$$

According to the procedures of the integrated algorithm shown in Fig. 4, the values of m_{left} and $m_{payload}$ are approximated from the analytical method proposed in Section 3.2 firstly. As shown in Fig. 6, the range of mass of propellant left for reboot is selected from 8 kg to 15 kg. In this range, the positive correlation and negative correlation in two powered phases are illustrated by a magenta line and a green line respectively. An intersection of the two line can be found and the values at this intersection are $m_{payload} = 148.70$ kg and $m_{left} = 11.91$ kg.

To further evaluate the accuracy of this analytical approach, the numerical investigation is employed to generate the correlation of $m_{payload}$ over the range of m_{left} for both powered phases. By solving the optimization problems of both powered phases over the range of m_{left}, optimized results including m_{left} and $m_{payload}$ can be obtained. Thus, the corresponding fitting curves can be generated by linear least-squares approximation respectively, which is shown in Fig. 7. By putting these two fitting curves together, an intersection between 1st- and 2nd-phase correlation lines indicates the real maximum mass of payload, which can be easily figured out in Fig. 8. The green and blue lines show the negative correlation in 2nd powered phase and pink and red show positive in 1st phase, and both the intersections yielded from numerical investigation and analytical derivation are marked in Fig. 9 as Intersection 1 and 2 respectively. The difference in the value of m_{left} is within 0.1 kg and the difference in the value of $m_{payload}$ is about 0.5 kg, i.e., no more than 0.34%. Such differences are staying in convergence range of the outer level of proposed optimization.

After finishing the following procedure to generate the initial combination of design variables using the values of m_{left} and $m_{payload}$ at the intersection, the integrated optimization of an ascent trajectory to the target SSO is performed. The optimal design variables are listed in Table 2 and corresponding curves of ballistic parameters are shown in Figs. 10 and 11. In particular, Fig. 10 provides a detailed description of the optimal ascent trajectory to the target SSO. Starting from the launch site, the launch vehicle finally sends its payload into the 500 km retrograde orbit. The transfer period can be obviously discovered from the succes-

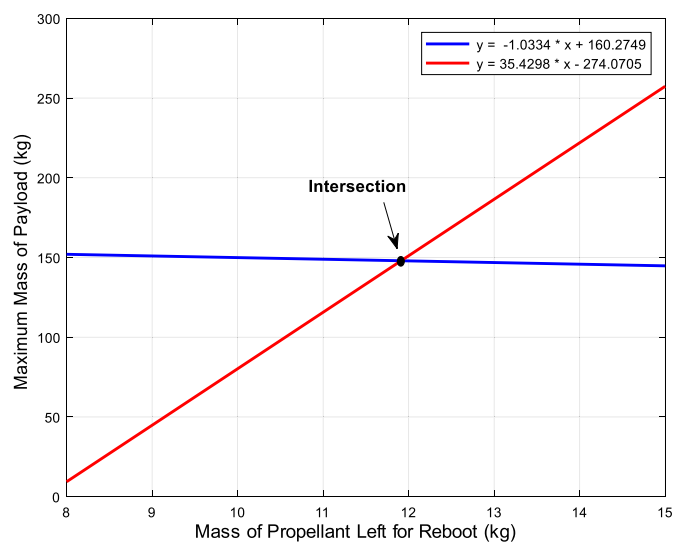

Fig. 8. Intersection of the two fitting curves in Fig. 6.

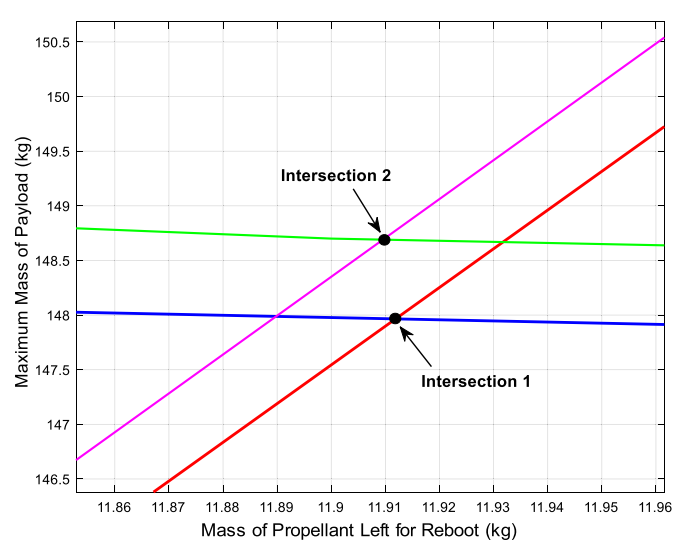

Fig. 9. Intersections generated from analytical method and numerical method.

Table 1a
Parameters of Electron rocket.

Specification	Stage 1	Stage 2
Length [m]	12.1	2.4
Diameter [m]	1.2	1.2
Dry Mass [kg]	815	275
Propellant Mass [kg]	9385	2050
Material/Structure	Carbon Fiber Composite	Carbon Fiber Composite
Propulsion	9×Rutherford Engines (LOX/Kerosene)	1×Rutherford Engine (LOX/Kerosene)
Lift-Off Thrust [kN]	154	22
Specific Impulse (Sea Level) [s]	252	333
Fairing	Diameter: 1.2 m, Length: 2.5 m, Mass: 50 kg	
Vehicle Mass (without payload) [kg]	12550	
Maximum Aerodynamic Pressure [MPa]	0.05	
Maximum Axial Acceleration	6 gravitational accelerations	

Table 1b
Parameters of Rutherford engine.

First Stage	Thrust (Sea Level) [kN]	17.05
	Specific Impulse (Sea Level) [s]	252
	Mass Flow Rate [kg/s]	6.902
Upper Stage	Thrust (Vacuum) [kN]	22.24
	Specific Impulse (Vacuum) [s]	333
	Mass Flow Rate [kg/s]	6.811

Table 1c
Parameters of target SSO and transfer orbit.

	Altitude of Perigee [km]	Altitude of Apogee [km]	Eccentricity	Inclination [deg]
Target SSO	500	500	10^{-6}	97.37478
Transfer Orbit	180	500	0.02384	97.37478

sive altitude profile as shown in Fig. 10a. According to Fig. 10b, the velocity of the vehicle increases steadily from start-off to the first time of SECO, and during the coast arc, the vehicle's velocity decreases from position k_1 to position k_2. The velocity of the vehicle experiences a short-term boost and consequently fulfills a successful injection of payload. A necessary criterion of successful injection is the zero-flight path angle at the injection positions. According to the successive curve of flight path angle in Fig. 10c, the values of flight path angle at two injection positions, k_1 and k_3, are approximately equal to zero, with a magnitude of 1×10^{-3} deg. Meanwhile, as shown in Fig. 10d, the pitch angle varies as the pitch law designed in Section 2.2. The optimal maximum mass of payload is 147.911 kg. According to the reference payload mass for the same target SSO provided by Payload User's Guide, the maximum mass of payload is 150 kg, which makes the simulation results more reliable.

4. Launching capability of electron rocket

In order to check the feasibility of the hierarchical ballistic design strategy, simulations based on the Electron rocket are executed. The parameters of the Electron rocket and the Rutherford engine are determined from the information in Payload User's Guide and listed in Table 1. And the global optimization method used in this case is GA. The target orbit is a standard SSO with an altitude of 500 km and a coplanar elliptic orbit with 180 km perigee is selected as the transfer orbit. The time of launch is selected at 4:00 a.m., September 1st, 2017 (UTC). The launch

Table 2
Optimal parameters of the ascent trajectory to SSO.

Launch Azimuth A_L [deg]	Maximum of Attack Angle α_M [deg]	Coefficient of Pitch-over κ
192.3381	0.3660	0.38525
Change of Rate of Pitch Angle in the First Part $\dot{\varphi}_1$ [deg/s]	**Change of rate of Pitch Angle in the Second Part $\dot{\varphi}_2$ [deg/s]**	**Change of Rate of Pitch Angle in the Third Part $\dot{\varphi}_3$ [deg/s]**
0.0052686	0.1749757	0.1605350
Time of Reboot of Second-stage Engine t_{reboot} [s]	**Mass of Propellant Left for Reboot m_{left} [kg]**	**Maximum Mass of Payload $m_{payload}$ [kg]**
3216.125	12.01	147.911

Table 3
Time consumption by different optimization algorithms.

Algorithm	Method in this paper	GA	PSO	DE
Time Consumption/min	13.56	313.2	635.6	1376.6

site is selected at Jiuquan Satellite Launch Center (40°58′03″N, 100°16′43″E), Gansu, China.

In addition, the parameters related with process constraints also verify the feasibility of the optimal design variables. As shown in Fig. 11, the maximum attack angle is no more than 0.5 degree and varies exponentially in a limited time span. The maximum of axial over-load is no more than 6 gravitational accelerations, which appears at the ending of each boost period. The maximum aerodynamic pressure encountered by the rocket is no more than 30 kPa, which is far below the prescribed aerodynamic pressure.

Compared with traditional optimization methods as GA, PSO and DE solving the optimization formulized in Eq. (5) with their initial guess generated randomly, the proposed two-level algorithm can achieve the same optimal results but is more efficient with the help of analytical initial guess, with their comparisons in time consumption shown in Table 3.

- The computer is set as the CPU Inter Core i5-7300HQ 2.50 GHz, GPU Geforce GTX 1050, RAM 4G with Windows 10 operation system. PGA is performed based on the CUDA Toolkit 8.1 and Visual Studio 2010, and genetic algorithm without parallelization in MATLAB 2017.

The comparisons above show with the help of analytical initial guess, the time consumption has been reduced greatly to a few tenths. Therefore, the proposed two-level optimization is so efficient that it is available to implement search an ergodic relationship between the maximum mass of payload and all the possible injected orbits, shown in Fig. 12.

It is found that the maximum mass of payload apparently changes with injected orbits: the higher the altitude is the less the maximum mass of payload is, and the mass is decreased at a linear rate with altitude, i.e., −0.085 kg/km for Electron Rocket; the larger the inclination is the mass is less, and a series of contour line on altitude-inclination plane are near parallel with their rate of −0.065 km/deg. In a word, these systematic investigations on all the possible injected orbits give a full detail of Electron Rocket.

If the injected orbit is fixed as a SSO, the relationship between the maximum mass of payload and the only altitude is just a subset from the ergodic one, as shown in Fig. 13. The mass optimized in this paper is the same as the one provided by Launcher User's Guide [29], which indicates all the results yielded above are credible.

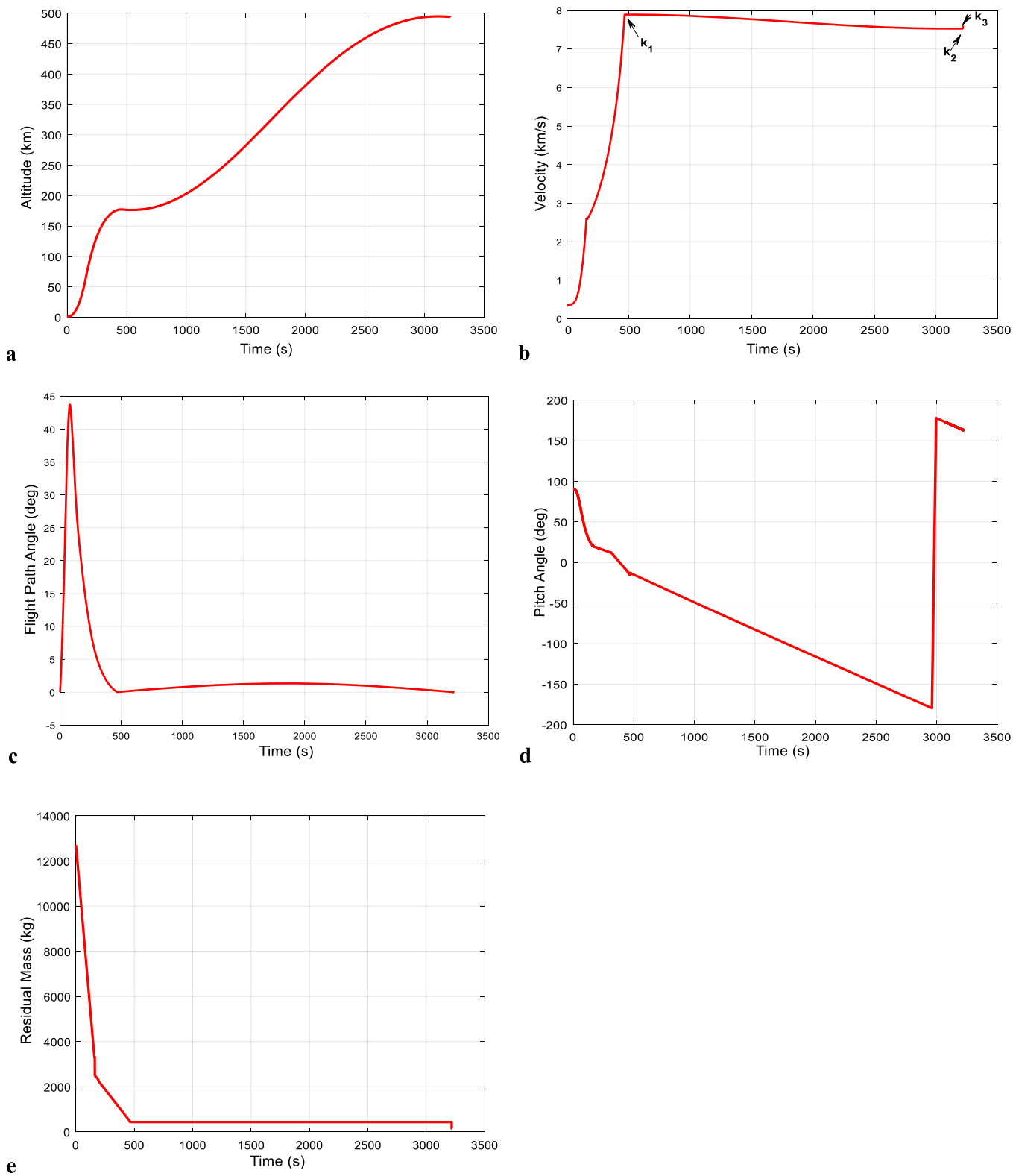

Fig. 10. Characteristic parameters of the optimal ascent trajectory to target SSO: **a** Successive altitude profile, **b** Successive velocity profile, **c** Successive flight path angle profile, **d** Successive pitch angle profile, **e** Successive residual mass profile.

5. Conclusion

An integrated algorithm using hierarchical optimization was proposed in this paper to design an optimal ascent trajectory to a target SSO for a small-lift launch vehicle. Equations of motion in the launch position reference frame were established and the pitch law of this specific flight was designed. After a general problem description of this trajectory design problem, nine design variables, three inequality constraints, and eight equality constraints were determined. An integrated optimization scheme using hierarchical optimization method was used to solve this problem. In consideration of efficiency, the relationship between two most sensitive

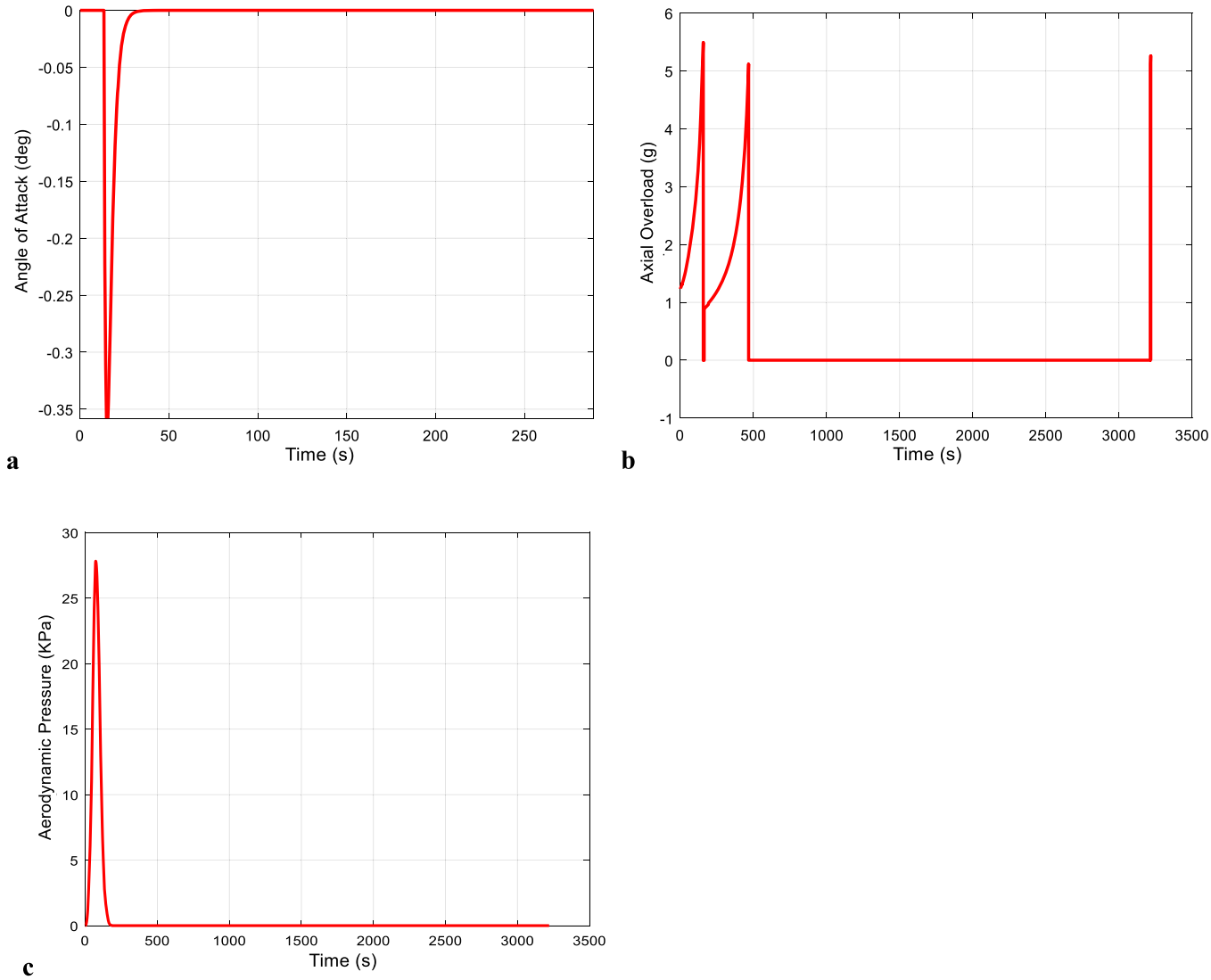

Fig. 11. Parameters related with process constraints: **a** Successive angel of attack profile, **b** Successive axial overload profile, **c** Successive aerodynamic pressure profile.

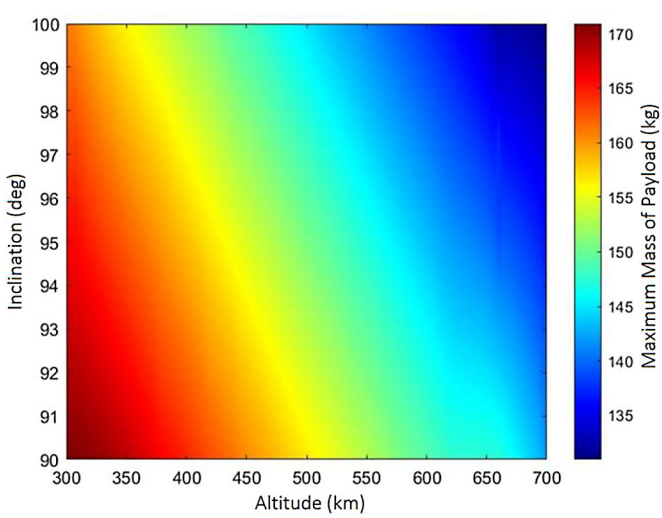

Fig. 12. Ergodic relationship between maximum mass of payload and injected orbits.

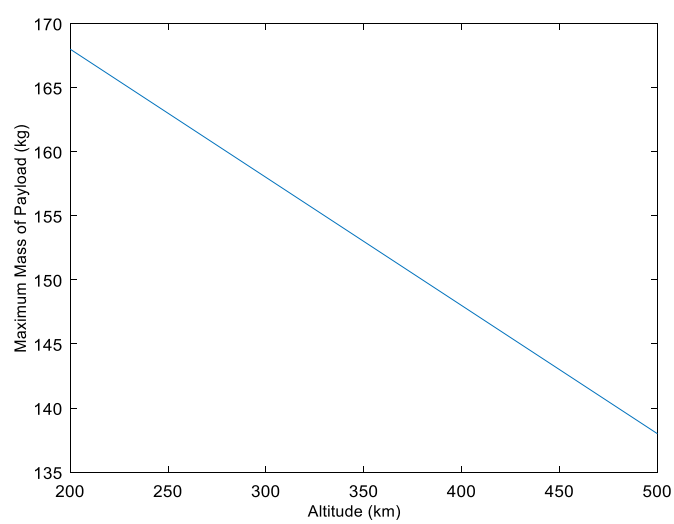

Fig. 13. Relationship between maximum mass of payload and orbital altitude of SSO.

design variables related with the coast arc was researched and an analytic approach was proposed to predetermine the two design variables, the mass of propellant left for reboot and the mass of payload. With the two-mass mentioned above, an initial combination of design variables was generated in the lower level by solving respectively two optimization problems of two powered phases.

Employing the starting values of design variables, the algorithm can perform a more efficient global optimization in higher level. Therefore, the algorithm gives an integrated two-level optimization algorithm for nine design variables of Electron Rocket with the initial payload and left mass are derived analytically based on the idea approximating burnout velocity. A simulation based on the Electron rocket and Rutherford engine was performed to evaluate the feasibility and efficiency of this algorithm. This integrated algorithm using hierarchical optimization was proved to be cost-effective and reliable after a detailed discussion of the generated ballistic parameters and a comparison with the reference data provided in Electron's Payload User's Guide.

The research in this paper offered future prospects for the analytic predetermination approach. Based on the empirical relationships, this analytic approach was in fact a utilization of the actual trajectories of existing launch vehicles. Thus, with the accumulation of ballistic data from the launch of more and more launch vehicles, the accuracy of this analytic approach was supposed to be further improved. In addition, such a method can be extended to more kinds of launch vehicles apart from the small-lift launch vehicle.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

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